A Structural Disproof of Non-Trivial Integer Cycle in Collatz Conjecture

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Abstract

The Collatz conjecture posits that any positive integer x, when repeatedly subjected to two operations— $\boldsymbol{3x+1}$ when $\boldsymbol{x}$ is odd and $\boldsymbol{x/2}$ when $\boldsymbol{x}$ is even—ultimately reaches 1 without exception. This paper presents a formal refutation of non-trivial cycles for the accelerated Collatz (Syracuse) operator. We establish an odd-to-odd iterative Diophantine equation and identify the trivial 1-cycle as an 'equilibrium state' where all division exponents $\boldsymbol{a_i = 2}$. Using an exhaustive structural partition, we prove that for any non-trivial integer cycle starting at $\boldsymbol{x > 1}$, there must exist a permutation of division exponents that deviates from this equilibrium. We demonstrate through Diophantine magnitude bounds and 3-adic valuation analysis that any such deviation triggers a terminal mathematical contradiction. The entire proof—comprising the exhaustive perturbation space and the resulting magnitude collapses—has been formally verified in Lean 4 using only foundational Mathlib axioms.

Keywords: Collatz conjecture, 3x+1 problem, Automated theorem proving, Lean 4, Diophantine equations, 3-adic valuation

MSC Classification: 11B83 , 03B35 , 11D61

1 Introduction

The Collatz conjecture, stated by Lothar Collatz in 1937, concerns the behavior of the function $f\colon \mathbb{Z}^+ \to \mathbb{Z}^+$ defined as:

$$f(n) = \begin{cases} 3n+1 & \text{if } n \equiv 1 \pmod 2 \\ \frac{n}{2} & \text{if } n \equiv 0 \pmod 2 \end{cases}$$

The conjecture asserts that for any starting integer n, the sequence of iterates $f^k(n)$ will eventually reach the integer 1. Despite its elementary statement, a formal proof has remained one of the most elusive problems in mathematics [5].

1.1 Notable prior works

1. The study of non-trivial integer cycles in the Collatz conjecture is centered on a fundamental Diophantine balance between the growth factor 3^r and the decay factor 2^n. In 1977, Ray Steiner [8] provided the first major breakthrough by rigorously excluding the existence of non-trivial "1-cycles" (circuits with a single local minimum) in the positive integers. By applying Baker's theorem on linear forms in logarithms, Steiner proved that the ratio of rising to falling steps must satisfy extremely rare rational approximations of $\log_2 3$, which are impossible for non-trivial integers.
2. Research on generalized 'm-cycles'—periodic orbits characterized by m local minima—was established by John Simons and Benne de Weger [7]. Using transcendence theory and Rhin's estimates for linear forms in logarithms, they proved that any hypothetical non-trivial cycle must possess a large number of 'peaks' and 'valleys'. Their work initially confirmed the non-existence of non-trivial cycles for $m \leq 68$ (later updated to $m \leq 75$), and recent refinements by Christian Hercher [3] have extended this barrier to $m \geq 92$.

3. Shalom Eliahou [2] introduced a criterion based on parity vectors and the "ones-ratio" theorem, showing that the fraction of odd terms in a cycle must be a convergent of the continued fraction expansion of $\log_2 3$. This mathematical framework allows researchers to translate computational verification limits into rigorous lower bounds for cycle length. Eliahou's method established that even at modest verification thresholds, any non-trivial cycle would be forced to contain millions or billions of elements.
4. High-performance computational projects, most notably those by David Barina [1], have verified the conjecture for all integers up to 2^{71} (approximately 2.36×10^{21}). This exhaustive verification implies that a non-trivial cycle, if it exists, must have a total period length exceeding $355,504,839,929$ steps. Furthermore, recent 2025 research by Jochen Kiemes [4] leveraging bit-position dynamics has derived a substantial structural lower bound of at least $17,026,679,261$ steps for any hypothetical cycle.
5. While non-trivial cycles are excluded in the positive domain up to these massive limits, the negative integer domain contains three confirmed non-trivial cycles at -1, -5, and -17. These known loops serve as critical benchmarks for contemporary structural proofs, such as Lionel Laurore's 2025 modulo 12 analysis [6]. These modern approaches attempt to demonstrate that deterministic residue class transitions preclude any additional periodicities, suggesting that the trivial $\{4, 2, 1\}$ cycle is indeed the only attractor for all positive integers.

1.1.1 Summary of lemmas for disproving non-trivial integer cycle

Theoretical Overview:

1. Lemma 0 (Valuation Distribution): Establishes the strict algebraic conditions for 2-adic valuation distribution, proving that adding a multiple of 2^a preserves base divisibility, and excluding higher divisibility relies strictly on parity.

2. Section 3 (Iterative Expansion): Defines the accelerated odd-to-odd operator $T(x) = (3x+1)/2^a$ and iterates it to construct the multi-step numerator sequence Z_n:

$$2^v \cdot k_{n+1} + m_{n+1} = \frac{3^n \cdot 2^v \cdot k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{a_1+\cdots+a_{n-1}}}{2^{a_1+a_2+\cdots+a_n}}$$

3. The Fundamental Cycle Equation: Mathematically locks the requirement for a periodic loop by equating the terminal iterated value to the initial value, yielding $(2^{S_n} - 3^n)x = Z_n$.
4. Lemma 1A & The Equilibrium State: Evaluates the baseline trajectory where every division exponent $a_i = 2$. Proves this state forces the Diophantine solution $x = 1$, isolating the trivial loop.
5. The Perturbation Model: Introduces the index-specific deviation metric $\delta_i = a_i - 2$ and the cumulative deviation S'_j to mathematically quantify trajectories that depart from the equilibrium baseline.
6. Lemma 1B (Exact Difference Formula): Formulates the exact rational difference between a perturbed numerator sum and the equilibrium numerator sum (ΔN_{actual}), scaling each term by the perturbation factor $(2^{S'_j} - 1)$.
7. Lemma 1C (Pure Negative Configuration): Evaluates trajectories containing only non-positive perturbations ($\delta_i \leq 0$). Proves the resulting magnitude deficit mathematically forces a strict inequality where the denominator is positive but bounded against the actual numerator.
8. Lemma 1D (Pure Positive Configuration): Evaluates trajectories containing only non-negative perturbations ($\delta_i \geq 0$). Directly proves that exponential accumulation forces the new denominator to outgrow the new numerator ($N_{\text{new}} < D_{\text{new}}$), forcing $x < 1$.

9. Lemma 1E (The Valuation Drop): Shifts the pure negative analysis to 3-adic arithmetic. Proves that the first instance of a negative perturbation ($\delta_r = -1$) introduces an irrecoverable 3-adic valuation deficit, locking the mathematical remainder to $v_3 = n - (r + 1)$.
10. Lemma 1F (Mixed Pair -1, +2): Evaluates a localized mixed perturbation state. Proves that an immediate +2 spike following a -1 drop fails to correct the Diophantine bounds, resulting in a strict magnitude collapse.
11. Lemma 1G (Minimal Mixed Pair -1, +1): Evaluates the minimal possible repair. Demonstrates that even the smallest positive perturbation inserted to offset a drop evaluates to an unrecoverable numerator collapse.
12. Lemma 1H (Exponential Repair Cost): Generalizes the mixed perturbation model. Employs the Lifting The Exponent (LTE) lemma to prove that repairing a 3-adic valuation deficit requires an exponentially escalating distance between the drop and the positive spike, strictly outgrowing the bounds of the cycle.
13. Lemma 2A: Establishes equivalence between the Collatz cycle and Diophantine identity.
14. Theorem 1 (Synthesis): Aggregates the exhaustive structural partition. With the equilibrium forcing $x = 1$, and all deviations (Pure Positive, Pure Negative, and Mixed) mathematically annihilated, the non-trivial cycle is unconditionally refuted.

2 Theory

2.1 Modular class analysis under Collatz operations

There are 2^n residue classes modulo 2^n. Since exactly half of the integers in $\{0, 1, \ldots, 2^n - 1\}$ are odd, the number of odd residue classes modulo 2^n is 2^{n-1}.

The odd modular classes can be written as: $2^n \cdot k + m$ (core integer $k \in \mathbb{Z}^+$, and residue $m \in \{1, 3, 5, \ldots, 2^n - 1\}$).

Let $N(a,n)$ be the number of odd residue classes m modulo 2^n such that the 2-adic valuation $v_2(3m+1)$ is exactly a. For positive integers $a \leq n$, this number is given by:

Lemma 0. *Let $N(a,n)$ be the number of odd residue classes m modulo 2^n such that the 2-adic valuation $v_2(3m+1)$ is exactly a. For positive integers $a \leq n$, this number is given by:*

$$N(a,n) = \begin{cases} 2^{n-a-1} & \text{if } 1 \leq a < n \\ 1 & \text{if } a = n \quad (\text{Edge case } v_2 \geq n) \end{cases}$$

Proof Let $N = 2^n \cdot k + m$, then $3N + 1 = 3 \cdot 2^n \cdot k + 3m + 1$.

$3 \cdot 2^n$ is divisible by 2^n.

Step I: Existence and Uniqueness: Since $\gcd(3, 2^a) = 1$, the congruence $3m + 1 \equiv 0 \pmod{2^a}$ has a unique solution modulo 2^a. Let this solution be $m \equiv c \pmod{2^a}$.

Step II: Lifting solution to mod 2^n:

Any solution $m \equiv c \pmod{2^a}$ can be expressed as $m = c + 2^a \cdot k$ for some integer k. To find all solutions modulo 2^n, we consider $k \pmod{2^{n-a}}$. There are 2^{n-a} distinct values of $k \pmod{2^n}$ leading to 2^{n-a} residues $m \pmod{2^n}$.

Step III: Excluding higher divisibility:

To ensure $3m + 1 \equiv 0 \pmod{2^n}$, we exclude solutions where $m \equiv c \pmod{2^n}$. Each solution modulo 2^n splits into two cases modulo 2^{a+1}:

a) One satisfies $3m + 1 \equiv 0 \pmod{2^{a+1}}$

b) The other does not satisfy $3m + 1 \equiv 0 \pmod{2^{a+1}}$

One half of the lifted solutions, i.e., 2^{n-a-1} remain valid for exact divisibility 2^a.

Step IV: Counting for each:

For $1 \leq a \leq n$: The number of residues $m \pmod{2^n}$ with $v_2(3m + 1) = a$ is 2^{n-a-1}.

Special edge case for $a = n$: There exactly 1 residue $m \pmod{2^n}$ such that $(3m + 1) \equiv 0 \pmod{2^n}$. □

Corollary 0-A. *According to Lemma 0, applying the $3x + 1$ operation to the set of odd modular classes yields the following distribution of division exponents:*

- 2^{n-2} results have divisibility by 2^1 (division exponent $a = 1$)
- 2^{n-3} results have divisibility by 2^2 (division exponent $a = 2$)
- 2^{n-4} results have divisibility by 2^3 (division exponent $a = 3$)
- 2^{n-5} results have divisibility by 2^4 (division exponent $a = 4$)

  $\vdots$
- $2^{n-n} = 1$ result has divisibility by 2^n (division exponent $a = n$)

The highest possible value of the division exponent $a = v_2(3x+1)$ also depends on the parity of k in the expression $3 \cdot 2^n \cdot k + 3m + 1$. After factoring out 2^n, the term becomes $3k + \frac{3m+1}{2^n}$. Since $3m + 1$ is always even, the parity of the entire expression depends on whether $3k$ is even or odd — which is determined by the parity of k. If k is odd, then $3k$ is also odd, and adding it to the (odd) quotient $\frac{3m+1}{2^n}$ yields an even result. Thus, the parity of k influences whether an additional factor of 2 can be extracted, affecting the total valuation $v_2(3x + 1)$. Therefore, the largest exponent (say, a_k) modulo 2^v is k-dependent such that $a_k \geq v$.

3 Multiple iterations on modular classes

We define the specific Collatz iteration terms as follows:

(a) I^1 represents a single $3x + 1$ operation on a starting odd integer N_0.

(b) I^2 represents division by 2^a, where $a = v_2(3N_0 + 1)$.

(c) I^d represents a single set of I^1 followed by an I^2 operation.

Let us assume a sequence with n allowed transformations in which the divisional exponents involved are $a_1, a_2, a_3, \ldots, a_n$.

The operation I^d applied to $2^v \cdot k_1 + m_1$ yields successive odd integers as follows:

Step 1:

$$2^v k_1 + m_1 \to \frac{3 \cdot 2^v \cdot k_1 + 3m_1 + 1}{2^{a_1}}$$

Step 2:

$$\frac{3 \cdot 2^v k_1 + 3m_1 + 1}{2^{a_1}} \to \frac{3^2 \cdot 2^v \cdot k_1 + 3^2 \cdot m_1 + 3 + 2^{a_1}}{2^{a_1+a_2}}$$

Step 3:

$$\cdots \to \frac{3^3 2^v k_1 + 3^3 \cdot m_1 + 3^2 + 3 \cdot 2^{a_1} + 2^{a_1+a_2}}{2^{a_1+a_2+a_3}}$$

Step 4:

$$\cdots \to \frac{3^4 2^v \cdot k_1 + 3^4 \cdot m_1 + 3^3 + 3^2 \cdot 2^{a_1} + 3 \cdot 2^{a_1+a_2} + 2^{a_1+a_2+a_3}}{2^{a_1+a_2+a_3+a_4}}$$

Step 5:

$$\cdots \to \frac{3^5 \cdot 2^v \cdot k_1 + 3^5 \cdot m_1 + 3^4 + \cdots + 3 \cdot 2^{a_1+a_2+a_3} + 2^{a_1+\cdots+a_4}}{2^{a_1+\cdots+a_5}}$$

For such a sequence with n steps ($1 \leq n \leq 2^{v-1}$), if the integer becomes $2^v \cdot k_{n+1} + m_{n+1}$, then we arrive at the **Fundamental Linear Diophantine Equation** of the Collatz conjecture:

$$2^v \cdot k_{n+1} + m_{n+1} = \frac{3^n \cdot 2^v \cdot k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{a_1+\cdots+a_{n-1}}}{2^{a_1+a_2+\cdots+a_n}}$$

Now, if $2^v \cdot k_{n+1} + m_{n+1} = 2^v \cdot k_1 + m_1$ (i.e., an integer reappears in a Collatz sequence after arbitrary n iterations of I^d), then by the equation above:

$$2^v \cdot k_1 + m_1 = \frac{3^n \cdot 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{a_1+\cdots+a_{n-1}}}{2^{a_1+a_2+\cdots+a_n}}$$

This is the fundamental linear Diophantine equation representing an integer loop.

Lemma 1A. *If all division exponents $a_i = p$, the only possible integer cycle is $4 \to 2 \to 1 \to 4$ when $p = 2$.*

Proof **Notation and setup**:

- Let n be the length of a hypothetical cycle, an integer where $n \geq 1$.
- Let $a = (a_1, \ldots, a_n)$ be the vector of division exponents, where each $a_i \in \mathbb{Z}^+$.
- Let $S_j = \sum_{i=1}^{j} a_i$ be the cumulative sum of the first j exponents, with the convention that $S_0 = 0$.
- Let $S = \sum_{i=1}^{n} a_i$ be the total sum of all exponents.
- The Numerator (N) of the cycle ratio is defined as: $N = N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$.
- The Denominator (D) of the cycle ratio is defined as: $D = 2^S - 3^n$.
- The Cycle Ratio (R) is defined as $R = \frac{N}{D}$.
- The Equilibrium State is defined as the special arrangement when all division exponents $a_i = 2$, $N = N_{\text{eq}}$, $D = D_{\text{eq}}$ and $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$. This implies $R_{\text{eq}} = 1$.

Recalling the fundamental equation, writing $S = \sum_{i=1}^{n} a_i$, and $S_j = \sum_{i=1}^{n-1} a_i$:

$$2^v \cdot k_1 + m_1 = \frac{3^n 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j}}{2^S}$$

Rearranging terms:

$$\begin{aligned} 2^v \cdot k_1 \cdot 2^S + m_1 \cdot 2^S = 3^n 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} \\ + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j} \end{aligned}$$

Grouping the k_1 and m_1 terms:

$$2^v \cdot k_1 (2^S - 3^n) = m_1 \cdot (3^n - 2^S) + 3^{n-1}$$

$$+ 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j}$$

Solving for k_1:

$$\begin{aligned} k_1 &= \frac{m_1(3^n - 2^S) + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j}}{(2^S - 3^n)} \cdot \frac{1}{2^v} \\ &= -\frac{m_1}{2^v} + \frac{3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j}}{2^v \cdot (2^S - 3^n)} \\ &= \frac{1}{2^v}[R - m_1] \end{aligned}$$

Where the ratio R is defined as:

$$R = \frac{3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \cdots + 2^{S_j}}{(2^S - 3^n)} = \frac{N}{D}$$

Applying the Equilibrium Condition: The lemma considers the equilibrium state, where all division exponents are equal to a constant integer p, i.e., $a_i = p$ for all $i = 1, \ldots, n$. Under this condition, the cumulative sums S_j take a simple, regular form:

- $S_0 = 0$
- $S_1 = p$
- $S_2 = p + p = 2p$
- In general, for any $j \geq 0$, the cumulative sum is $S_j = j \cdot p$.

Formulating the Numerator Sum: Substituting this into the general formula for the numerator gives the equilibrium numerator, N_{eq}:

$$N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{j \cdot p}.$$

Demonstrating the Geometric Progression: To show that this sum is a Geometric Progression (GP), we write out the first few terms:

- For $j = 0$: The term is $3^{n-1} \cdot 2^{0 \cdot p} = 3^{n-1}$.
- For $j = 1$: The term is $3^{n-2} \cdot 2^{1 \cdot p} = 3^{n-2} \cdot 2^p$.
- For $j = 2$: The term is $3^{n-3} \cdot 2^{2 \cdot p} = 3^{n-3} \cdot (2^p)^2$.

The sum can be written as: $N_{\text{eq}} = 3^{n-1} + 3^{n-2} \cdot 2^p + 3^{n-3} \cdot (2^p)^2 + \ldots + 3^0 \cdot (2^p)^{n-1}$.

Considering any j-th term, $T_j = 3^{n-1-j} \cdot (2^p)^j$, the ratio of consecutive terms is constant:

$$\frac{T_{j+1}}{T_j} = \frac{2^p}{3}$$

Since this ratio is constant for all terms, the sum is a Geometric Progression with the following parameters:

- **First Term (A)**: 3^{n-1}
- **Common Ratio (r)**: $\left(\frac{2^p}{3}\right)$
- **Number of Terms**: n

Summing the Series and Finalizing the Proof:

We apply the standard formula for the sum of a finite geometric series, sum $= A \cdot \left[\frac{r^n - 1}{r-1}\right]$:

$$N_{\text{eq}} = 3^{n-1} \cdot \left[\frac{\left(\frac{2^p}{3}\right)^n - 1}{\frac{2^p}{3} - 1}\right] \implies N_{\text{eq}} = \frac{2^{np} - 3^n}{2^p - 3}$$

The equilibrium denominator is $D_{\text{eq}} = 2^S - 3^n = 2^{np} - 3^n$. The cycle ratio R is therefore:

$$R = \frac{N}{D} = \left[\frac{2^{n \cdot p} - 3^n}{2^p - 3}\right] \cdot \left[\frac{1}{2^{n \cdot p} - 3^n}\right] \Rightarrow R = \frac{1}{2^p - 3}$$

For R to be a positive integer, the denominator $2^p - 3$ must equal 1. This occurs only when $2^p = 4$, which implies $p = 2$. In this case, $R = 1$. Substituting into the equation for the core integer, $k_1 = \frac{1}{2^v} \cdot [R - m_1]$, gives $k_1 = \frac{1}{2^v} \cdot [1 - m_1]$. Since k_1 must be a non-negative integer and m_1 is a positive odd residue, the only solution is $m_1 = 1$, which yields a valid $k_1 = 0$. This corresponds to the integer 1, proving that the only possible cycle with uniform exponents is the trivial one. □

Corollary 1A-1: The Threshold Ratio. *For R to be a positive integer, the denominator $D = 2^S - 3^n$ must be greater than 0. The threshold value for which $2^S - 3^n > 0$ is $2^S > 3^n$, or $\frac{S}{n} > \log_2 3$. This gives us the threshold ratio:*

$$\frac{S}{n} > 1.585, \quad \text{or} \quad S > 1.585 \cdot n$$

4 The Perturbation Model

By Lemma 1A, it is evident that no integer solution is possible for R unless all division exponents $a_i = 2$. Assuming this equilibrium, where $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$, as a base case, all possible combinations of division exponents can be examined by incrementing or decrementing them from this equilibrium (base) position. The model assumes that a non-trivial cycle is formed by some appropriate combination of division exponents that makes $R = \frac{N}{D}$ a positive integer. The following perturbations cover all valid division exponent combinations:

- **Case I:** Pure increment (positive perturbations) of the exponents from a_1 to a_n.
- **Case II:** Pure decrement (negative perturbations) of the division exponents from a_1 to a_n.
- **Case III:** Mixed perturbations with net increment in division exponent sum.
- **Case IV:** Mixed perturbations with net decrement in division exponent sum.
- **Case V:** Mixed perturbations with no net alteration in the division exponent sum (total increment = total decrement).

All cases can be captured in two scenarios: either an increase or a decrease in the numerator's magnitude while keeping $N_{\text{new}} > D_{\text{new}}$. There is, however, another possibility in which, by some exceptional coincidence, $N_{\text{new}} = N_{\text{eq}}$ is achieved via a net decrement of the total division exponents.

Lemma 1B (Exact Difference Formula). *Due to valid perturbations of division exponents from equilibrium, the possible outcomes are: (a) $N_{new} > N_{eq}$, (b) $N_{new} < N_{eq}$, or (c) a non-trivial $N_{new} = N_{eq}$.*

Proof **Notation and Setup**:

- Let $n \geq 1$ be a fixed integer representing the cycle length.
- Let $a = (a_1, \ldots, a_n)$ be the exponent vector, where each $a_i \geq 1$.
- Let $\delta_i = a_i - 2$ be the perturbation of each exponent from the equilibrium value of 2.

- Let $S'_j = \sum_{i=1}^{j} \delta_i$ be the prefix sums of the perturbations, with $S'_0 = 0$.
- Let $s_j = \sum_{i=1}^{j} a_i$ be the cumulative sum of the exponents ($s_j = 2j + S'_j$).
- Let $N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{s_j}$ be the perturbed numerator.
- Let N_{eq} be the equilibrium numerator corresponding to $a_{\text{eq}} = (2, 2, \ldots, 2)$.

The proof establishes an exact identity for the numerator's deviation and categorizes its outcomes.

1. **Derivation of the Exact Difference Formula**: We begin with the definition of N_{new} and substitute $s_j = 2j + S'_j$:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j+S'_j} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot 2^{S'_j}$$

The equilibrium numerator N_{eq} is the case where all $\delta_i = 0$ (implying $S'_j = 0$ and $2^{S'_j} = 1$):

$$N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j}$$

Subtracting N_{eq} from N_{new} and factoring out the common term $3^{n-1-j} \cdot 2^{2j}$ yields the identity:

$$\begin{aligned} N_{\text{new}} - N_{\text{eq}} &= \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot 2^{S'_j} - \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \\ &= \sum_{j=1}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot \left[2^{S'_j} - 1\right] \end{aligned}$$

2. **Consequences of the Identity**: Let the sensitivity weights be defined as $w_j = 3^{n-1-j} \cdot 2^{2j}$, which are strictly positive for all j. The identity $\Delta N = \sum w_j [2^{S'_j} - 1]$ leads to three exhaustive logical possibilities:

- **Equality Case ($N_{\mathbf{new}} = N_{\mathbf{eq}}$)**: Occurs if and only if $S'_j = 0$ for all j. A sum of terms that are all ≥ 0 or all ≤ 0 can only be zero if every term is zero. Thus, $\delta_i = 0$ for all i.
- **Monotone Sign Cases**: (a) If $S'_j \geq 0$ for all j and at least one $S'_j > 0$, then $N_{\text{new}} > N_{\text{eq}}$. Each factor $(2^{S'_j} - 1)$ is non-negative, and at least one is strictly positive. (b) If $S'_j \leq 0$ for all j and at least one $S'_j < 0$, then $N_{\text{new}} < N_{\text{eq}}$. Each term is non-positive, and at least one is strictly negative.
- **Mixed-Sign Prefixes**: For general non-trivial perturbations, the relationship evaluates to either $N_{\text{new}} > N_{\text{eq}}$, $N_{\text{new}} < N_{\text{eq}}$, or a coincidental $N_{\text{new}} = N_{\text{eq}}$.

Conclusion: Any valid perturbation routes exactly into one of these definitive magnitude outcomes relative to the equilibrium state. □

4.1 Constraints in the Perturbation Model

There are mathematical constraints that create a deterministic frame around this model:

Constraint 1: Division Exponent Lower Bound. The division exponent is defined as $a_i = v_2(3x_{i-1}+1)$. We note that since x_{i-1} is a positive odd integer, $(3x_{i-1}+1)$ is a positive even integer, which implies the constraint that $a_i \geq 1$ for all i. Consequently, the only possible negative perturbation from the equilibrium value of 2 is $\delta_i = -1$, i.e., **no division exponent can be lowered more than once**.

Constraint 2: Modulo 3 Incompatibility. Both D and N at any form (perturbed or unperturbed) are odd and not divisible by 3.

Constraint 3: Ratio Bounds. If an integer cycle forms, the ratio $Z = \frac{N_{\text{new}}}{D_{\text{new}}} \geq 5$ for all $N_{\text{new}} > D_{\text{new}}$ cases, and $\frac{N_{\text{new}}}{D_{\text{new}}} = 1$ for some non-trivial $N_{\text{new}} = D_{\text{new}}$.

Lemma 1C (Pure Negative Configuration). *Let us assume $n \geq 1$ and let the exponent-vector $a = (a_1, \ldots, a_n)$ satisfy $1 \leq a_i \leq 2$ for every i. Let us also assume*

that the perturbation is non-trivial, meaning at least one $a_i < 2$ (and thus $a_i = 1$). Let $S = \sum_{i=1}^{n} a_i$ be the total sum of exponents. The negatively perturbed numerator and denominator are defined as: $N_{new} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S_j}$ and $D_{new} = 2^S - 3^n$. Under these conditions, it is always true that $N_{new} > D_{new}$.

Proof **Notation and Setup**:

- Let p be the number of negative perturbations (i.e., exponents $a_i = 1$). By the premise, $p \geq 1$.
- The total sum of the exponents S is exactly $S = 2n - p$.

The proof proceeds by establishing a tight lower bound for the numerator, N_{new}, and then showing that this lower bound is strictly greater than the denominator, D_{new}.

1. **Establishing a Lower Bound for the Cumulative Sums (S_j)**: For any index j in the range $0 \leq j \leq n-1$, the portion of the total sum of exponents *after* index j is $S - S_j = \sum_{i=j+1}^{n} a_i$. Since each $a_i \leq 2$, this sum is bounded above by the number of terms multiplied by 2:

$$S - S_j = \sum_{i=j+1}^{n} a_i \leq 2(n-j)$$

   Rearranging this inequality gives a lower bound for the cumulative sum S_j:

$$S_j \geq S - 2(n-j)$$

2. **Establishing a Lower Bound for the Numerator (N_{new})**: We insert this lower bound for S_j into the definition of the numerator:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S_j} \geq \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S-2(n-j)}$$

Factoring out the term 2^S and introducing $m = n - j$ gives:

$$N_{\text{new}} \geq 2^S \cdot \sum_{m=1}^{n} 3^{m-1} \cdot 4^{-m}$$

The sum component is a Geometric Progression with $A = 1$, $r = 3/4$, and n terms. Summing the series gives $1 - (3/4)^n$. Substituting this back yields the lower bound for N_{new}:

$$N_{\text{new}} \geq 2^S \left[1 - \left(\frac{3}{4}\right)^n\right]$$

3. **Comparing the Numerator and Denominator**: We analyze the difference $N_{\text{new}} - D_{\text{new}}$ using this lower bound:

$$\begin{aligned} N_{\text{new}} - D_{\text{new}} &\geq 2^S \cdot \left[1 - \left(\frac{3}{4}\right)^n\right] - (2^S - 3^n) \\ &\geq 2^S - 2^S \left(\frac{3}{4}\right)^n - 2^S + 3^n \\ &\geq 3^n - \left(\frac{3}{4}\right)^n \cdot 2^S \end{aligned}$$

4. **Finalizing the Inequality**: We substitute the exact value for $S = 2n - p$:

$$N_{\text{new}} - D_{\text{new}} \geq 3^n - \left(\frac{3}{4}\right)^n \cdot 2^{2n-p}$$

Since $4^n = 2^{2n}$, we can simplify:

$$N_{\text{new}} - D_{\text{new}} \geq 3^n - \left(\frac{3^n}{2^{2n}}\right) \cdot 2^{2n} \cdot 2^{-p} = 3^n - 3^n \cdot 2^{-p}$$

Factoring out 3^n gives the explicit numeric bound:

$$N_{\text{new}} - D_{\text{new}} \geq 3^n \left(1 - \frac{1}{2^p}\right)$$

Conclusion: Since $p \geq 1$, the term $\frac{1}{2^p}$ is always positive and less than or equal to $\frac{1}{2}$. Therefore, the term $(1 - \frac{1}{2^p})$ is always ≥ 0.5. As 3^n is also strictly positive, their product is strictly greater than 0. This formally proves that $N_{\text{new}} > D_{\text{new}}$. □

Lemma 1D (Pure Positive Configuration). *Any purely positive perturbation of the division exponents from the equilibrium leads to a non-integer ratio $\frac{N_{new}}{D_{new}}$, violating the cycle condition.*

Proof **Notation and Setup**:

- $a = (a_1, \ldots, a_n)$: The vector of perturbed division exponents.
- $\delta_i = a_i - 2$: The perturbation for each exponent. For this lemma, all $\delta_i \geq 0$, and at least one $\delta_i > 0$.
- $S'_j = \sum_{i=1}^{j} \delta_i$: The prefix sum of the first j perturbations. By convention, $S'_0 = 0$.
- $S'_n = \sum_{i=1}^{n} \delta_i$: The total sum of all perturbations. By the lemma's premise, $S'_n \geq 1$.
- $N_{\text{eq}}, D_{\text{eq}}$: The numerator and denominator at the equilibrium state ($a_i = 2$). We know $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$.
- $N_{\text{new}}, D_{\text{new}}$: The perturbed numerator and denominator.
- $\Delta N = N_{\text{new}} - N_{\text{eq}}$ and $\Delta D = D_{\text{new}} - D_{\text{eq}}$: The net change in the numerator and denominator.

The proof proceeds in four steps. We will show that the increase in the denominator (ΔD) is strictly greater than the increase in the numerator (ΔN).

1. **The Objective**: Our goal is to prove that $D_{\text{new}} > N_{\text{new}}$. Since $D_{\text{eq}} = N_{\text{eq}}$ at equilibrium, this is equivalent to proving that the change in the denominator is strictly greater than the change in the numerator ($\Delta D > \Delta N$).
2. **Calculating the Exact Change in the Denominator (ΔD)**: The total sum of the perturbed exponents is $S = 2n + S'_n$. The new denominator is $D_{\text{new}} = 2^S - 3^n =$

$2^{2n+S'_n} - 3^n$. We calculate ΔD directly:

$$\begin{aligned}\Delta D &= D_{\text{new}} - D_{\text{eq}} \\ &= (2^{2n+S'_n} - 3^n) - (2^{2n} - 3^n) \\ &= 2^{2n} \cdot 2^{S'_n} - 2^{2n}\end{aligned}$$

Factoring out the 2^{2n} term gives the exact change:

$$\Delta D = 2^{2n}\left(2^{S'_n} - 1\right)$$

3. **Establishing a Strict Upper Bound for ΔN**: We use the Exact Difference Formula from Lemma 1B:

$$\Delta N = \sum_{j=1}^{n-1}\left[3^{n-1-j}2^{2j}\left(2^{S'_j} - 1\right)\right]$$

Since this is a purely positive perturbation ($\delta_i \geq 0$), the sequence of prefix sums is non-decreasing: $0 = S'_0 \leq S'_1 \leq \ldots \leq S'_{n-1} \leq S'_n$. This means that for any prefix sum S'_j, $S'_j \leq S'_n$. Consequently:

$$\left(2^{S'_j} - 1\right) \leq \left(2^{S'_n} - 1\right)$$

We establish an upper bound for ΔN by replacing each term in the sum with this maximum factor:

$$\Delta N \leq \sum_{j=1}^{n-1}\left[3^{n-1-j}2^{2j}\left(2^{S'_n} - 1\right)\right]$$

Factoring out the constant $\left(2^{S'_n} - 1\right)$ yields the exact definition of the equilibrium numerator, N_{eq}. This gives us the strict upper bound:

$$\Delta N \leq \left(2^{S'_n} - 1\right) \cdot N_{\text{eq}}$$

4. **Comparison**: By definition, $N_{\text{eq}} = 2^{2n} - 3^n$. Since $n \geq 1$, $3^n > 0$, which implies $N_{\text{eq}} < 2^{2n}$. Since the lemma requires at least one positive perturbation, $S'_n \geq 1$, making $(2^{S'_n} - 1)$ a positive number. It follows directly that:

$$\left(2^{S'_n} - 1\right) \cdot N_{\text{eq}} < \left(2^{S'_n} - 1\right) \cdot 2^{2n}$$

Substituting our terms back, we get $\Delta N < \Delta D$.

Conclusion: Since the increase in the denominator is strictly greater than the increase in the numerator, it is proven that $D_{\text{new}} > N_{\text{new}}$. Therefore, no purely positive perturbation can satisfy the condition for an integer cycle. □

Constraint 4: Formulation of Actual ΔN. The sum formula of the perturbed numerator is:

$$N_{\text{new}} = 3^{n-1} + 3^{n-2}2^{S_1} + 3^{n-3}2^{S_2} + \cdots + 2^{S_{n-1}}$$

where $S_1, S_2, \ldots, S_{n-1}$ are the perturbed prefix sums of exponents.

The unperturbed (equilibrium) numerator is:

$$N_{\text{eq}} = 3^{n-1} + 3^{n-2} \cdot 2^2 + 3^{n-3} \cdot 2^4 + \cdots + 2^{2n-2}$$

If perturbations cause $N_{\text{eq}} > N_{\text{new}}$, then $\Delta N_{\text{actual}} = N_{\text{eq}} - N_{\text{new}}$. In this difference, all unperturbed terms cancel out, leaving only the sum of differences at perturbed positions. If a perturbation is introduced at the k-th term, the actual change is

$\Delta N_{\text{actual}} = \sum_{k=2}^{n}(\text{Term}_{\text{eq},k} - \text{Term}_{\text{new},k})$, specifically:

$$\text{Term}_k = 3^{n-k} \cdot \left[2^{2k-2} - 2^{S_{k-1}}\right]$$

Conversely, if $N_{\text{new}} > N_{\text{eq}}$, then $\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}}$, and the general term becomes:

$$\text{Term}_k = 3^{n-k} \cdot \left[2^{S_{k-1}} - 2^{2k-2}\right]$$

Constraint 5: Bounds on Negative Perturbations. The number of pure negative perturbations p must satisfy $p \geq 1$ and $p < 0.415 \cdot n$. This is derived from the threshold ratio $S > 1.585n$. Since $S = 2n - p$ (in the pure negative case), we have:

$$2n - p > 1.585n \implies p < (2 - 1.585)n \implies p < 0.415n$$

Similarly, any total decrement in a mixed case must maintain this bound to ensure $D_{\text{new}} > 0$.

Constraint 6: Formulation of Required ΔN. This equation defines the exact value of the net change in the numerator, $\Delta N_{\text{required}}$, required for a cycle with a specific integer ratio $Z \geq 5$ to exist.

1. Derivation for Net Decrement ($N_{\text{new}} < N_{\text{eq}}$): Start with the cycle condition $Z = \frac{N_{\text{new}}}{D_{\text{new}}}$, which implies $N_{\text{new}} = Z \cdot D_{\text{new}}$. Substituting $N_{\text{new}} = N_{\text{eq}} - \Delta N$:

$$\begin{aligned}
N_{\text{eq}} - \Delta N &= Z \cdot D_{\text{new}} \\
\Delta N &= N_{\text{eq}} - Z \cdot D_{\text{new}} \\
&= (2^{2n} - 3^n) - Z(2^S - 3^n) \\
&= (2^{2n} - Z \cdot 2^S) + (Z - 1)3^n
\end{aligned}$$

This yields the final equation for net decrement:

$$\Delta N_{\text{required}} = (2^{2n} - Z \cdot 2^S) + (Z - 1)3^n \tag{1}$$

2. Derivation for Net Increment ($N_{\mathbf{new}} > N_{\mathbf{eq}}$): Substituting $N_{\text{new}} = N_{\text{eq}} + \Delta N$:

$$\begin{aligned} N_{\text{eq}} + \Delta N &= Z \cdot D_{\text{new}} \\ \Delta N &= Z(2^S - 3^n) - (2^{2n} - 3^n) \\ &= (Z \cdot 2^S - 2^{2n}) + (1 - Z)3^n \end{aligned}$$

This yields the final equation for net increment:

$$\Delta N_{\text{required}} = (Z \cdot 2^S - 2^{2n}) + (1 - Z)3^n \tag{2}$$

The Fundamental Cycle Condition. According to the perturbation model, an integer cycle is formed if and only if the actual change in the numerator, ΔN_{actual}, reconciles perfectly with the theoretically required change, $\Delta N_{\text{required}}$. Thus, the fundamental equation to be tested for the existence of any cycle is:

$$\boxed{\Delta N_{\text{required}} = \Delta N_{\text{actual}}}$$

5 Mathematical Consequences of Perturbations

Lemma 1E (First Negative-Perturbation Lemma). *Let a perturbation from equilibrium contain at least one negative component ($\delta_i = -1$). Let r be the index of the first such negative perturbation in the sequence. Then the corresponding term T_{r+1}*

within the sum for ΔN_{actual} has a 3-adic valuation given by $v_3(T_{r+1}) = n - (r+1)$. Since $r \geq 1$, this valuation is strictly positive and strictly less than n [$0 < v_3(T_{r+1}) < n$].

Proof **Notation and Setup**:

- Let $a_1, a_2, \ldots, a_n$ be a sequence of integers.
- Let $r \geq 1$ be the first index for which $a_r \neq 2$. We consider the specific case where this first perturbation is $a_r = 1$. This implies that $a_1 = a_2 = \ldots = a_{r-1} = 2$.
- Let $S_k = \sum_{i=1}^{k} a_i$ denote the partial sum of the sequence.
- For the index $k = r+1$, we define a quantity $d_k = |S_{k-1} - (2k-2)|$.

The proof proceeds by verifying that d_k is an odd integer and then using this fact to calculate the valuation.

1. **Calculating the Partial Sum at the Point of Perturbation**: By our hypothesis, the sequence begins with $r-1$ terms equal to 2, followed by the term $a_r = 1$. The partial sum up to index r, denoted S_r, is therefore:

$$S_r = \sum_{i=1}^{r} a_i = 2(r-1) + 1 = 2r - 1$$

2. **Proof that d_k is Odd**: The lemma defines d_k for $k = r+1$. Note that the definition uses the partial sum S_{k-1}, which becomes $S_{r+1-1} = S_r$.

$$\begin{aligned} d_{r+1} &= |S_r - (2(r+1) - 2)| \\ &= |S_r - 2r| \end{aligned}$$

Substituting the value of S_r from Step 1:

$$d_{r+1} = |(2r-1) - 2r| = |-1| = 1$$

Since $d_{r+1} = 1$, it is an odd integer.

3. **Calculating the 3-adic Valuation**: We now analyze the expression for $k = r+1$. First, we algebraically manipulate the term inside the valuation by factoring out the common power of 2:

$$\begin{aligned} 3^{n-k} \cdot \left\{2^{S_{k-1}} - 2^{2k-2}\right\} &= 3^{n-k} \cdot 2^{2k-2} \cdot \left\{2^{S_{k-1}-(2k-2)} - 1\right\} \\ &= 3^{n-k} \cdot 2^{2k-2} \cdot \{2^{d_k} - 1\} \end{aligned}$$

Using the additive property of valuations, $v_p(xyz) = v_p(x) + v_p(y) + v_p(z)$, we evaluate each term:

- $v_3(3^{n-k}) = n - k$, by the definition of 3-adic valuation.
- $v_3(2^{2k-2}) = 0$, because 3 is not a prime factor of any power of 2.
- For the third term, we established in Step 2 that d_k is odd. For any odd integer d_k, $2^{d_k} - 1$ is never divisible by 3. Modulo 3, $2 \equiv -1$, so $2^{d_k} \equiv (-1)^{d_k} \equiv -1 \pmod 3$. Thus, $2^{d_k} - 1 \equiv 1 \pmod 3$, meaning its 3-adic valuation is 0.

Combining these results, the total 3-adic valuation is $(n - k) + 0 + 0 = n - k$.

4. **Establishing the Upper Bound**: By definition, $r \geq 1$. The lemma sets $k = r+1$. Therefore, the smallest possible value for k is 2. If $k \geq 2$, then $-k \leq -2$. Adding n to both sides gives $n - k \leq n - 2$.

5. **Application of the Non-Archimedean Strong Triangle Inequality**: The 3-adic valuation function v_3 satisfies the ultrametric inequality $v_3(x + y) \geq \min(v_3(x), v_3(y))$, with strict equality $v_3(x+y) = \min(v_3(x), v_3(y))$ if $v_3(x) \neq v_3(y)$. The actual change in the numerator is the sum $\Delta N_{\text{actual}} = \sum T_j$. Since T_{r+1} is the first non-zero deviation term, it introduces a strict valuation of $n - (r+1) < n$. By the strong triangle inequality, the total valuation $v_3(\Delta N_{\text{actual}})$ is strictly bounded to exactly $n-(r+1)$ unless there exists at least one subsequent term T_m $(m > r+1)$ such that $v_3(T_m) \leq n - (r + 1)$.

Conclusion: This confirms that the initial negative perturbation introduces a term with a strictly positive valuation exactly equal to $n-(r+1) < n$. Under the non-Archimedean strong triangle property, this establishes an irrecoverable arithmetic deficit that mathematically dictates the valuation of the entire sum, forcing $v_3(\Delta N_{\text{actual}}) < n$ unless explicitly repaired by subsequent exponent perturbations. □

Corollary 1E-1. *The valuation drop established in Lemma 1E implies that for the 3-adic valuation of the entire sum ΔN_{actual} to be n or greater, it is necessary that at least one subsequent term T_k is generated by a positive perturbation that causes its corresponding exponent difference, d_k, to be a non-zero even integer.*

Proof Lemma 1E proves that the term T_{r+1} has a 3-adic valuation strictly less than n. For the valuation of the entire sum $v_3(\Delta N_{\text{actual}})$ to be raised to n or higher, compensation is required from another term T_k (where $k > r+1$). The valuation of any such term is $v_3(T_k) = n-k+v_3(2^{d_k}-1)$. An increase in this valuation can only come from the $v_3(2^{d_k}-1)$ component being positive (≥ 1), which requires the congruence $2^{d_k} \equiv 1 \pmod 3$. This holds if and only if the exponent $d_k = |S_{k-1} - (2k-2)|$ is a non-zero even integer, which in turn necessitates positive perturbations. **Conclusion**: The valuation drop can only be compensated for if a subsequent positive perturbation creates a term T_k where the exponent difference d_k is a non-zero even integer. This establishes the necessary condition analyzed in Lemma 1H. □

Lemma 1F (Mixed Pair -1, +2). *If a local perturbation from the equilibrium vector $(2, 2, \ldots, 2)$ at index k ($0 \leq k \leq n-1$) replaces the pair $(a_k, a_{k+1}) = (2, 2)$ with $(a'_k, a'_{k+1}) = (1, 4)$, the growth of the denominator surpasses the growth of the numerator, i.e., $N_{new} < D_{new}$.*

Proof **Notation and Setup**:

- Let the equilibrium state be defined by $a_{\text{eq}} = (2, 2, \ldots, 2)$, where $S_j^{\text{eq}} = 2j$ and $S^{\text{eq}} = 2n$.
- The equilibrium numerator and denominator are $N_{\text{eq}} = D_{\text{eq}} = 4^n - 3^n$.

The proof proceeds by deriving an exact expression for the difference $D_{\text{new}} - N_{\text{new}}$ to demonstrate a strict magnitude collapse.

1. **State Parameters after Perturbation**: The perturbation changes the total sum of exponents by $(-1) + (+2) = +1$. The new total sum is $S_{\text{new}} = 2n + 1$. The new denominator is $D_{\text{new}} = 2^{2n+1} - 3^n$. This gives us the exact change in denominator:

$$\Delta D = D_{\text{new}} - D_{\text{eq}} = (2^{2n+1} - 3^n) - (2^{2n} - 3^n) = 2^{2n}$$

2. **The Change in the Numerator (ΔN)**: We calculate the change $\Delta N = N_{\text{new}} - N_{\text{eq}}$.

   - For $j < k$, the partial sums S_j are unchanged ($\Delta T_j = 0$).
   - For $j = k$, the partial sum S_k decreases by 1. This halves the term T_k, resulting in a change of $\Delta T_k = -\frac{1}{2}T_k$.
   - For $j \geq k + 1$, the partial sum S_j increases by 1. This doubles the term T_j, resulting in a change of $\Delta T_j = T_j$.

   Summing these effects yields the total change:

$$\Delta N = -\frac{1}{2}T_k + \sum_{j=k+1}^{n-1} T_j$$

   Normalizing this by dividing by $\Delta D = 2^{2n}$, and letting $M = n-1-k$, the geometric series evaluates to:

$$\frac{\Delta N}{2^{2n}} = 1 - \frac{9}{8}\left(\frac{3}{4}\right)^M$$

   Hence, $\Delta N = 2^{2n}\left[1 - \frac{9}{8}\left(\frac{3}{4}\right)^M\right]$.

3. **The Difference $D_{\text{new}} - N_{\text{new}}$**: We calculate the final difference:

$$D_{\text{new}} - N_{\text{new}} = (D_{\text{eq}} + \Delta D) - (N_{\text{eq}} + \Delta N)$$

$$
\begin{aligned}
&= \Delta D - \Delta N \quad (\text{since } N_{\text{eq}} = D_{\text{eq}}) \\
&= 2^{2n} - 2^{2n}\left[1 - \frac{9}{8}\left(\frac{3}{4}\right)^M\right] \\
&= 2^{2n} \cdot \frac{9}{8}\left(\frac{3}{4}\right)^M
\end{aligned}
$$

Conclusion: Since all terms in the final expression are strictly positive, $D_{\text{new}} - N_{\text{new}} > 0$. Any negative perturbation immediately proceeded by a $+2$ perturbation violates the cycle condition by forcing $D_{\text{new}} > N_{\text{new}}$. □

Lemma 1G (Minimal Mixed Pair -1, +1). *If a local perturbation from the equilibrium vector* $(2, 2, \ldots, 2)$ *at index* k *(*$0 \leq k \leq n-1$*) replaces the pair* $(a_k, a_{k+1}) = (2, 2)$ *with* $(a'_k, a'_{k+1}) = (1, 3)$*, this forces* $N_{new} < D_{new}$*.*

Proof **Notation and Setup**:

- We introduce a mixed perturbation at an arbitrary position k: $\delta_k = -1$ and $\delta_{k+1} = +1$.
- All other exponents remain at their equilibrium value: $a_i = 2$ for $i \notin \{k, k+1\}$.

1. **Calculating the Change in the Denominator (ΔD)**: The net change in the sum of exponents is $\Delta S = -1 + 1 = 0$. The new sum of exponents is $S_{\text{new}} = 2n$. Therefore, the new denominator is identical to the equilibrium denominator ($D_{\text{new}} = D_{\text{eq}}$). The change in the denominator is $\Delta D = 0$.
2. **Calculating the Change in the Numerator (ΔN)**: Using the difference identity $\Delta N = \sum 3^{n-1-j} \cdot 2^{2j}\left[2^{S'_j} - 1\right]$, we calculate the prefix sums S'_j:
   - For $j < k$: All δ_i are zero, so $S'_j = 0$.
   - For $j = k$: $S'_k = \delta_k = -1$.
   - For $j > k$: $S'_j = \delta_k + \delta_{k+1} = 0$.

The prefix sum S'_j is non-zero **only** for the single index $j = k$. For all other indices, the term $\left[2^{S'_j} - 1\right]$ evaluates to 0. Therefore, the entire sum collapses to a single term:

$$\begin{aligned}\Delta N &= 3^{n-1-k} \cdot 2^{2k} \cdot \left[2^{-1} - 1\right] \\ &= -\frac{1}{2}\left[3^{n-1-k} \cdot 2^{2k}\right]\end{aligned}$$

Conclusion: We have proven that $\Delta D = 0$ while ΔN is a strictly negative value. Since $D_{\text{eq}} = N_{\text{eq}}$ at equilibrium, it follows that $(D_{\text{eq}} + \Delta D) > (N_{\text{eq}} + \Delta N)$, establishing $D_{\text{new}} > N_{\text{new}}$. □

Corollary 1G-1 (The Principle of Net Algebraic Cost). *The change in the denominator is given by* $\Delta D = 2^{2n}(2^{S'_n} - 1)$*, while the change in the numerator is* $\Delta N = \sum_{j=0}^{n-1}\left[3^{n-1-j}2^{2j}\left(2^{S'_j} - 1\right)\right]$*. Any mixed perturbation, regardless of its structure, is algebraically fatal.*

Proof Any mixed perturbation requires at least one negative perturbation $\delta_k = -1$. This guarantees that the sum ΔN_{mixed} contains negative-value terms. Consider a "purely positive" scheme that results in the same final exponent sum S'_n. The sum for ΔN_{pure} contains only positive terms. Therefore, the mixed sum is strictly less than the pure sum ($\Delta N_{\text{mixed}} < \Delta N_{\text{pure}}$). Lemma 1D has already proven that for any positive perturbation, the denominator growth exceeds the numerator growth ($\Delta D > \Delta N_{\text{pure}}$). By transitivity, $\Delta D > \Delta N_{\text{pure}} > \Delta N_{\text{mixed}}$. **Conclusion**: This proves that any mixed perturbation forces $D_{\text{new}} > N_{\text{new}}$, making an integer cycle impossible. □

Lemma 1H (Exponential Repair Cost). *If the* 3*-adic valuation of any term* T_k *in* ΔN_{actual} *is* $v_3\left[2^{d_k} - 1\right] = r$*, then* $d_k = 2 \cdot B \cdot 3^{r-1}$ *(where* $B \in \mathbb{Z}^+$ *and is not divisible by* 3*).*

Proof The proof proceeds in four steps. First, we establish the necessary parity of the exponent d_k. Second, we apply the Lifting The Exponent (LTE) Lemma. Third, we solve for the 3-adic structure of d_k. Finally, we combine these constraints.

1. **Condition for a Non-Zero Valuation**: For the valuation $v_3\left[2^{d_k}-1\right]$ to be a positive integer $r \geq 1$, it is necessary that 3 divides $(2^{d_k}-1)$. This is equivalent to $2^{d_k} \equiv 1 \pmod 3$. Since $2 \equiv -1 \pmod 3$, this becomes $(-1)^{d_k} \equiv 1 \pmod 3$, which holds if and only if d_k is an **even** integer. Thus, $d_k = 2m$.
2. **Application of the Lifting The Exponent (LTE) Lemma**: We rewrite the expression as $2^{2m}-1 = 4^m - 1^m$ and apply the LTE Lemma for the prime $p = 3$. The preconditions hold (3 is an odd prime, $3 \nmid 4$, $3 \nmid 1$, and $3 \mid (4-1)$). Applying the formula $v_p(x^n - y^n) = v_p(x-y) + v_p(n)$:

$$v_3(4^m - 1^m) = v_3(4-1) + v_3(m) \implies v_3\left[2^{d_k}-1\right] = 1 + v_3(m)$$

3. **Solving for the 3-adic Structure of d_k**: Setting our derived expression equal to r:

$$1 + v_3(m) = r \implies v_3(m) = r - 1$$

Substituting back $m = \frac{d_k}{2}$ and using $v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b)$:

$$v_3(d_k) - v_3(2) = r - 1$$

Since 2 is not divisible by 3, $v_3(2) = 0$. The equation simplifies to $v_3(d_k) = r - 1$. This means the highest power of 3 that divides d_k is exactly 3^{r-1}.
4. **Final Form of d_k**: d_k must be even, and $d_k = K \cdot 3^{r-1}$ where $3 \nmid K$. Since 3^{r-1} is always odd, K must be an even integer. Writing $K = 2B$, the condition $3 \nmid K$ becomes $3 \nmid 2B$, which simplifies to $3 \nmid B$. Since d_k must be positive, B must be positive.

Conclusion: Combining these facts gives the exact required form for $d_k = 2 \cdot B \cdot 3^{r-1}$, where $B \in \mathbb{Z}^+$ and $3 \nmid B$. $\square$

Corollary 1H-1 (The Escalating Cost of Delayed Compensation).

1. **The Initial Valuation Drop**: By **Lemma 1E**, a negative perturbation at index $k-1$ creates a term T_k in the sum ΔN_{actual} with a 3-adic valuation of $v_3(T_k) = n-k$.
2. **The Compensation Requirement**: For another term T_l $(l > k)$ to compensate for this, its valuation must be at least equal to the valuation of the term it is canceling. We assume a minimal requirement that $v_3(T_l) \geq n - k$.
3. **Quantifying the Requirement**: The valuation of any term T_l is given by $v_3(T_l) = n - l + v_3(2^{d_l} - 1)$. Substituting the requirement from Step 2:

$$n - l + v_3(2^{d_l} - 1) \geq n - k \implies v_3(2^{d_l} - 1) \geq l - k$$

4. **Applying Lemma 1H**: **Lemma 1H** proves that for the valuation $v_3(2^{d_l} - 1)$ to equal rank r, d_l must be of the form $d_l = 2 \cdot B \cdot 3^{r-1}$. To satisfy the inequality from Step 3, we must achieve a rank of at least $r = l - k$. Therefore, d_l must be of the form:

$$d_l = 2 \cdot B \cdot 3^{(l-k)-1}$$

Conclusion: The required power of 3 in the structure of the exponent difference d_l grows exponentially with the distance $l - k$ between the negative perturbation and its compensating positive perturbation. To construct an exponent difference divisible by such a high power of 3, the prefix sum of perturbations S'_{l-1} must be carefully engineered using a massive exponential set of positive perturbations.

6 Theorem 1: Formal Refutation of Non-Trivial Cycles

Theorem 6.1 (Refutation of Non-Trivial Cycles) *For the accelerated Collatz (Syracuse) operator defined as $T(x) = (3x+1)/2^{a_i}$, there exist no non-trivial integer cycles for any starting integer $x > 1$. The only permissible cyclic trajectory is the uniform equilibrium state ($a_i = 2$), which strictly forces $x = 1$.*

Proof **1. Declaration of Symbols and Notations**

- n: The length of a hypothetical non-trivial cycle.
- $a = (a_1, a_2, \ldots, a_n)$: The vector of division exponents, $a_i \geq 1$.
- $\delta_i = a_i - 2$: The perturbation of an exponent from the equilibrium value of 2.
- $S = \sum a_i$: The total sum of exponents.
- N_{new}, D_{new}: The perturbed numerator and denominator, where $D_{new} = 2^S - 3^n$.
- Z: The required integer ratio for a cycle, $Z = N_{new}/D_{new}$.
- $A = \Delta N_{actual}$: The actual change in the numerator.
- $B = Z \cdot 2^S - 2^{2n}$: The algebraic term derived from the denominator's change.
- $C = -3^n(Z-1)$: The 3-adic term derived from the cycle identity.

2. Proof by Contradiction: The Two Conditions We assume a hypothetical non-trivial Collatz cycle of length n. For this cycle to be mathematically possible, it must simultaneously satisfy two fundamental conditions:

1. **The Algebraic Condition:** The ratio $Z = N_{new}/D_{new}$ must be a positive integer. This necessitates that $N_{new} \geq D_{new}$.
2. **The Arithmetic Condition:** The cycle's parameters must satisfy the core Diophantine identity $A - B = C$. This requires equal 3-adic valuations: $v_3(A - B) = v_3(C)$.

The proof proceeds by an exhaustive structural partition of the perturbation vector δ.

3. Case 1: Pure Positive Perturbations (All $\delta_i \geq 0$)

- **Refutation:** Refuted by Lemma 1D.
- **Argument:** Lemma 1D proves that any purely positive perturbation forces the denominator to grow strictly faster than the numerator ($D_{new} > N_{new}$), making an integer ratio mathematically impossible.

4. Case 2: Pure Negative Perturbations (All $\delta_i \leq 0$)

- **Refutation:** Refuted by Lemma 1E.
- **Argument:** While this configuration satisfies the Algebraic Condition ($N_{new} > D_{new}$), Lemma 1E proves that the initial negative perturbation locks the total 3-adic valuation of the numerator change to a strict upper bound of $v_3 < n$. Without positive perturbations to enact Lemma 1H compensation, the Arithmetic Condition strictly fails.

5. Case 3: Mixed Perturbations (At least one $\delta_i > 0$ and one $\delta_i < 0$) This case is mathematically refuted by demonstrating that the required arithmetic repair cost strictly violates the available algebraic bounds.

- **The Algebraic Constraint:** To satisfy the Algebraic Condition ($N_{new} \geq D_{new}$), the total exponent sum S must be strictly bounded below exponential growth.
- **The Arithmetic Cost:** By Lemma 1E, negative perturbations introduce a 3-adic valuation drop. By Corollary 1H-1, repairing this drop requires a positive perturbation term at an index distance $l - k$. By the Lifting The Exponent (LTE) lemma, this required distance scales exponentially with the valuation deficit.
- **Synthesis (Magnitude Collapse):** The formal verification proves that the structural distance required for the 3-adic repair ($l - k$) strictly outgrows the bounded perturbation allowance. Satisfying the Arithmetic Condition mathematically forces the net exponent sum to strictly exceed the equilibrium bound ($S > 2n$).

- **Conclusion:** By Lemma 1D, any state where $S > 2n$ unconditionally forces $D_{new} > N_{new}$. The arithmetic requirements and algebraic boundaries are mutually exclusive.

6. Case 4: Non-Trivial Equilibrium ($Z = 1$) By Lemma 1A and 1B, a non-trivial trajectory possessing $N_{new} = N_{eq}$ and $Z = 1$ is analytically refuted.

7. Final Conclusion: All exhaustive structural states lead to strict contradictions. Pure Positive and Pure Negative cycles are impossible. Mixed cycles trigger a terminal magnitude collapse. Since no non-trivial state can exist without violating fundamental bounds, the trivial $1 \to 4 \to 2 \to 1$ loop is the exclusive integer cycle. The theorem is proven. □

7 Equivalence

Lemma 2A: Fundamental Equivalence (Dynamics $\iff$ Diophantine). *A sequence of positive odd integers forms an n-step Collatz integer cycle with a specific sequence of division exponents if and only if its starting integer is a solution to a corresponding linear Diophantine equation.*

Proof **1. Symbol Definitions**

- $x_0, x_1, \ldots, x_n$: A sequence of $n + 1$ positive odd integers representing the cycle's path.
- $a_1, a_2, \ldots, a_n$: A sequence of n positive integer division exponents, where each $a_i \geq 1$.
- S_j: The partial sum of the first j exponents, $S_j = \sum_{i=1}^{j} a_i$, with $S_0 = 0$. The total sum is $S = S_n$.
- T: An integer constant defined by the exponent sequence:

$$T = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$$

2. Proof

The proof is bidirectional. We first prove that a Collatz cycle implies the Diophantine identity (the forward direction) and then prove that the identity implies a valid Collatz cycle (the "backward" direction).

Part 1: Forward Direction (Collatz Cycle $\implies$ Diophantine Identity)

Hypothesis: We assume a valid n-step Collatz integer cycle exists. This means we have a sequence of positive odd integers starting with x_0 such that for each step $x_i = \frac{3x_{i-1}+1}{2^{a_i}}$ and the sequence loops back, so $x_n = x_0$.

Derivation: By iteratively substituting the formula for each x_i into the next, we derive a general formula for x_n in terms of only x_0:

$$x_n = \frac{3^n \cdot x_0 + T}{2^S}$$

Applying the cycle condition, $x_n = x_0$, and rearranging the terms yields the Diophantine identity:

$$(2^S - 3^n)x_0 = T$$

The forward direction is proven.

Part 2: Backward Direction (Diophantine Identity $\implies$ Collatz Cycle)

Hypothesis and Non-degeneracy: Suppose a positive odd integer x_0 and integers $a_1, \ldots, a_n \geq 1$ satisfy the Diophantine identity:

$$(2^S - 3^n)x_0 = T \quad (*)$$

First, we note that the identity is non-degenerate because $2^S \neq 3^n$ (as one is even and the other is odd), so the term $(2^S - 3^n)$ is a non-zero integer. We will construct integers $x_1, \ldots, x_n$ with $x_i = \frac{3x_{i-1}+1}{2^{a_i}}$ and show that $x_n = x_0$ by induction on n.

Base Step (First Division): We reduce the identity (*) modulo 2^{a_1}. Every term in the sum for T except the leading term (3^{n-1}) is divisible by 2^{a_1}. Thus:

$$(2^S - 3^n)x_0 \equiv 3^{n-1} \pmod{2^{a_1}}$$

Since $S \geq a_1$, this simplifies to $-3^n x_0 \equiv 3^{n-1} \pmod{2^{a_1}}$. As $\gcd(3, 2^{a_1}) = 1$, we can divide by 3^{n-1} to get:

$$-3x_0 \equiv 1 \pmod{2^{a_1}} \implies 3x_0 + 1 \equiv 0 \pmod{2^{a_1}}$$

This proves that $(3x_0 + 1)$ is divisible by 2^{a_1}, and the integer $x_1 = \frac{3 \cdot x_0 + 1}{2^{a_1}}$ is well-defined.

Exactness of Valuation: We must now prove that x_1 is odd, which requires proving that the valuation is exact: $a_1 = v_2(3x_0+1)$. If we assume for contradiction that the valuation is higher, $v_2(3x_0 + 1) \geq a_1 + 1$, it can be shown that this leads to a parity contradiction in the reduced form of the Diophantine identity. The explicit form of the reduced tail term $T^{(1)}$ is odd modulo 2, while the left-hand side would be even. This contradiction proves the valuation must be exact.

Inductive Reduction: By substituting x_1 back into the original identity (*) and simplifying, the equation reduces to the exact same Diophantine form for a cycle of length $n-1$ starting at x_1:

$$(2^{S-a_1} - 3^{n-1})x_1 = \sum_{j=1}^{n-1} 3^{n-1-j} 2^{S_j - a_1} = T^{(1)}$$

Thus, the pair $(x_1, (a_2, \ldots, a_n))$ satisfies the same form of identity for $n-1$ steps. By the same reduction argument applied iteratively, we obtain a sequence of valid Collatz transformations, and the final reduced identity after n steps is $(2^0 - 3^0)x_n = T^{(n)}$, which simplifies to $x_n = x_0$.

Conclusion

The forward construction shows every Collatz cycle produces a solution to the identity. The backward reduction proves every solution yields a valid Collatz cycle with the exact valuations $a_i = v_2(3x_{i-1} + 1)$. The correspondence is therefore a bijection. This proposition formally closes the equivalence gap and establishes the Diophantine equation as a faithful model for the analysis of integer cycles. $\square$

8 Conclusion

Theorem 1 formally proves that no trajectory of the accelerated Collatz operator can regenerate an integer except the trivial $1 \to 4 \to 2 \to 1$ cycle. Therefore, non-trivial integer cycles do not exist.

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- **Computational tools:** The formalization in this work was developed with the assistance of LLM-based developer tooling (e.g., syntax autocompletion and boilerplate reduction). These tools were utilized strictly to accelerate the encoding of the author's mathematical definitions into Lean 4 syntax. The mathematical logic, theorem statements, and proof strategy were manually constructed and verified by the author. The author assumes full responsibility for the content of the manuscript. The validity of the proof is certified by the Lean 4 kernel, not by the generative tool.

Code availability. The complete Lean 4 formalization code and automated verification scripts supporting the findings of this study have been submitted as electronic supplementary material for peer review. The source code is also publicly available in the associated GitHub repository at https://github.com/adityabagchi2002-boop/Non-Trivial-Cycle-Refutation-in-Collatz-Dynamics-. The code is axiom-free and relies strictly on the foundational Mathlib library.

References

[1] D. Barina, Convergence verification of the Collatz problem, *The Journal of Supercomputing* **77** (2021), 2681–2688.

[2] S. Eliahou, The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths, *Discrete Math.* **118** (1993), 45–56.

[3] C. Hercher, There are no Collatz m-Cycles with $m \leq 91$, *J. Integer Sequences*, **26** (2023), Article 23.3.5.

[4] J. Kiemes, Structural lower bounds for the Collatz cycle length via bit-position dynamics, arXiv preprint (2025).

[5] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *Amer. Math. Monthly* **92** (1985), 3–23.

[6] L. Laurore, Deterministic residue class transitions and modulo 12 analysis of the Collatz conjecture, arXiv preprint (2025).

[7] J. Simons and B. de Weger, Theoretical and computational bounds for m-cycles of the $3n + 1$ problem, *Acta Arith.* **117** (2005), 51–70.

[8] R. P. Steiner, A theorem on the Syracuse problem, *Proceedings of the 7th Manitoba Conference on Numerical Mathematics* (1977), 553–559.